Towards Context-Aware Image Anonymization with Multi-Agent Reasoning

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Abstract

Indirect, context-dependent PII

Street-level imagery captures indirect, context-dependent personally identifiable information (PII) (vehicle markings, signage, contextual objects) that Large Vision Language Models (LVLMs) exploit to infer private attributes at up to 76.4% accuracy [1] and to localize users at city-scale [2]. **Identified limitations in existing approaches**:

- **Over-process**: Gaussian blur drops CityScapes instance-seg. AP by 5.3% [3]; Still, 95.9% id. recovery from blurred faces in CelebA-HQ [4].
- **Miss context**: single-pass detectors (used in DP2 [5], FADM [6], SVIA [7]) cover fixed taxonomies; indirect PII is open-vocabulary.
- **No audit**: black-box pipelines lack reasoning trails.
- **Cloud APIs** conflict with data protection.

Research Question: "Can orchestrated agents achieve context-aware image anonymization while preserving data utility and enabling accountability?"

Contributions:

- **Agentic PDCA refinement.** (Fig. 1) 3 specialized agents coordinate anonymization via round-robin in bounded Plan–Do–Check–Act cycles.

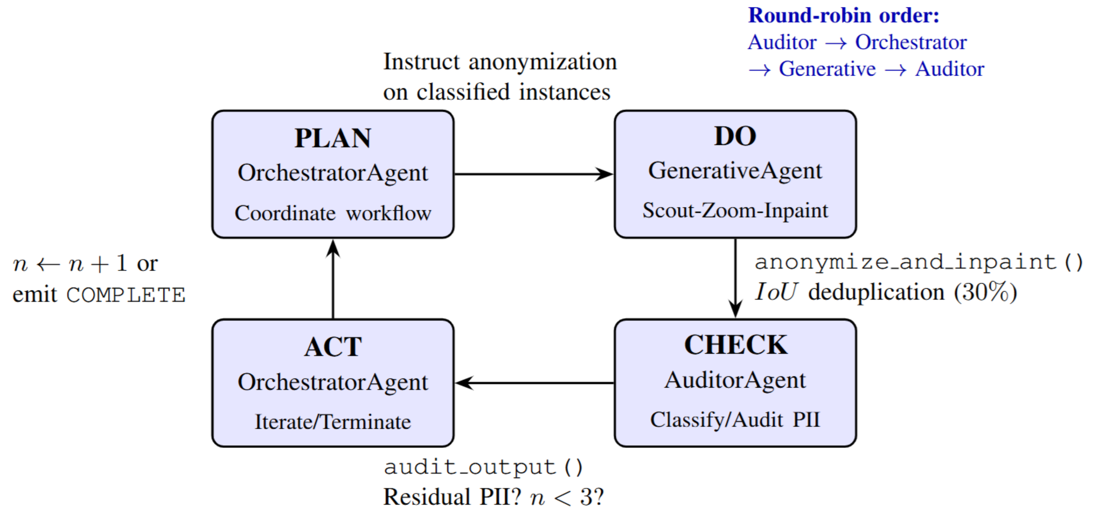

Fig. 1. Round-robin PDCA coordination with three specialized agents. Phase 1 applies deterministic preprocessing for direct PII. Phase 2 implements bounded iterative refinement. (© Deggendorf Institute of Technology)

- **On-premise audit trails.** Open-source models; structured logs supporting transparency.
- **Appearance decorrelation.** SDXL [8] + ControlNet [9] inpainting breaks person re-identification (Re-ID) vectors while preserving pose and scene utility.
- Failure modes (acknowledgment-without-execution, premature completion) mitigated via Orchestrator history validation and Auditor visual check.

Methods and Materials

Two-phase agentic anonymization pipeline

- **Phase 1: Pre-defined detectors.** YOLOv8-seg (**persons**), custom YOLOv8s on UC3M-LP [10] (**plates**), YOLO-TS [11] (**signs**). Pose-preserved SDXL+OpenPose [12] inpainting with prompt-conditioned randomized clothing palette to break appearance correlation.
- **Phase 2: Agentic. Round-robin AutoGen [13] group chat (Auditor → Orchestrator → Generative).** Qwen2.5-VL-32B [14] classifies **indirect PII**; Grounded-SAM-2 [15,16] segments scout-and-zoom crops; SDXL+Canny inpaints. Bounded with IoU deduplication and audit verification.

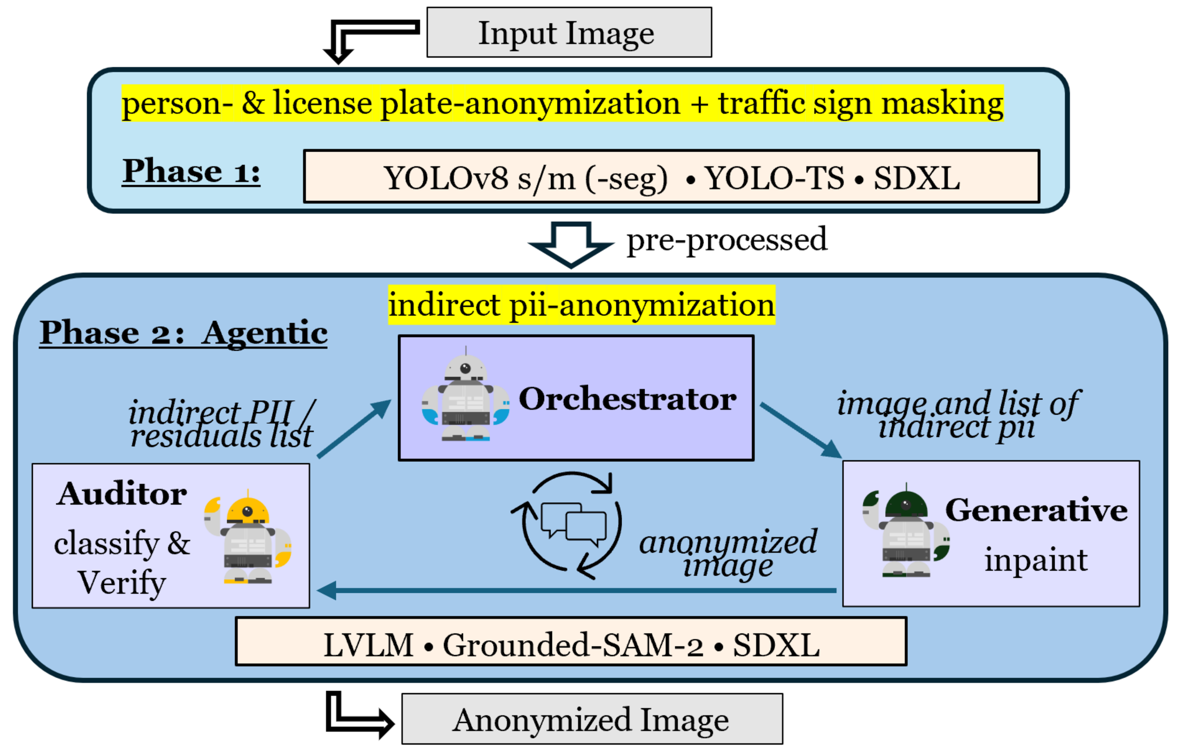

Fig. 2. Two-phase agentic anonymization architecture. (© Deggendorf Institute of Technology)

Results

Re-ID reduction and utility preservation

- **Image Quality** (Table 1 and 3). Achieves in-class distribution fidelity on CUHK03-NP [17] (FID 20.4, KID 0.014) (cf. Fig. 3) and CityScapes [18] (FID 9.1, LPIPS 0.025). Avoids the severe perceptual distortion of DP2, the drift of Gaussian Blur, and vastly outperforms SVIA.
- **Privacy-Utility Balance** (Table 2). Drops Re-ID risk (R1) on CUHK03-NP by **72.9%** (down to 16.9%). Unlike DP2 (which destroys distribution) and FADM (which retains too much identity), our method hits sweet spot.
- **Downstream Utility** (Table 4). Granular replacement preserves scene context. Maintains a highly accurate mIoU of 0.877 on CityScapes (vs. SVIA's 0.478), keeping static background classes near-perfect (cf. Fig. 4).

Results

Table 1. Image quality on CUHK03-NP (n=6,732). Bold = best. Lower better except SSIM.

Method	MSE ↓	SSIM ↑	LPIPS ↓	KID ↓	FID ↓
Gauss. Blur	1097.9	0.633	0.382	0.224	178.5
DP2 [5]	3275.1	0.443	0.303	0.066	59.7
FADM [6]	**446.4**	**0.785**	**0.157**	0.032	33.3
Ours	**975.6**	**0.699**	**0.203**	**0.014**	**20.4**

Table 2. Re-ID risk on CUHK03-NP. ResNet50 [19], original query vs. anonymized gallery + re-ranking. Lower = stronger privacy.

Method	mAP % ↓	R1 % ↓	R5 % ↓
Original	66.0	62.4	73.1
Gauss. Blur	6.4 *−90%*	9.4 *−85%*	15.1 *−79%*
DP2 [5]	**4.4** *−93%*	**8.6** *−86%*	**12.7** *−83%*
FADM [6]	32.9 *−50%*	33.4 *−46%*	52.9 *−28%*
Ours	**13.7** *−79%*	**16.9** *−73%*	**31.0** *−58%*

Table 3. Image quality on CityScapes (n=1,525). Lower better across all metrics.

Method	LPIPS ↓	KID ↓	FID ↓
SVIA [7]	0.530	0.027	44.3
Ours	**0.025**	**0.001**	**9.1**

Table 4. Downstream segmentation utility on CityScapes. SegFormer [20] pseudo-GT; prediction gap anonymizations vs. original. Higher

Class	mIoU (Ours) ↑	mIoU (SVIA) ↑
All	0.877	0.478
road	0.996	0.955
building	0.977	0.677
person	0.778	0.150
car	0.975	0.567
sky	0.995	0.820

Color key. Light blue = our method; red bold = best per metric; small italic numbers = % reduction from original dataset w/o anonymizations.

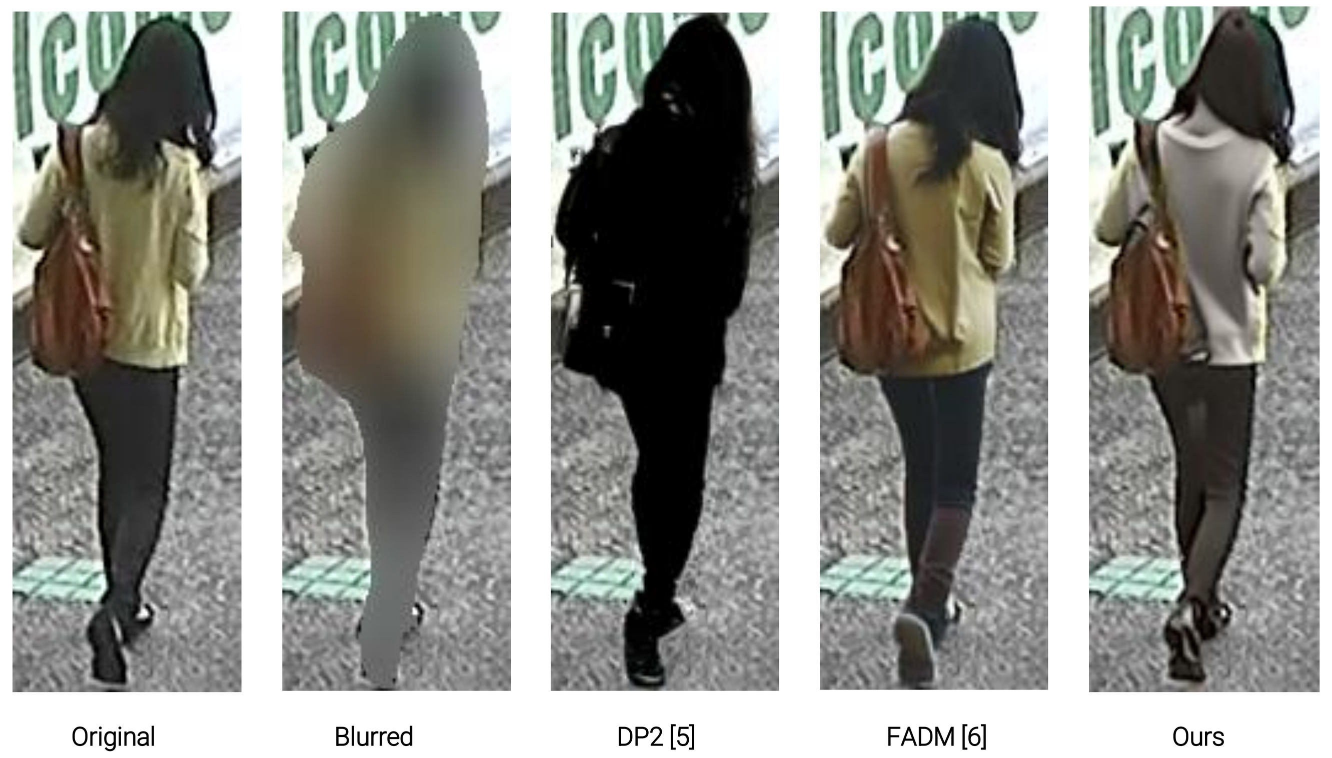

Fig. 3. Qualitative example: pose preserved, identity broken. Subject image © CUHK03 dataset [17], research use.

Discussion

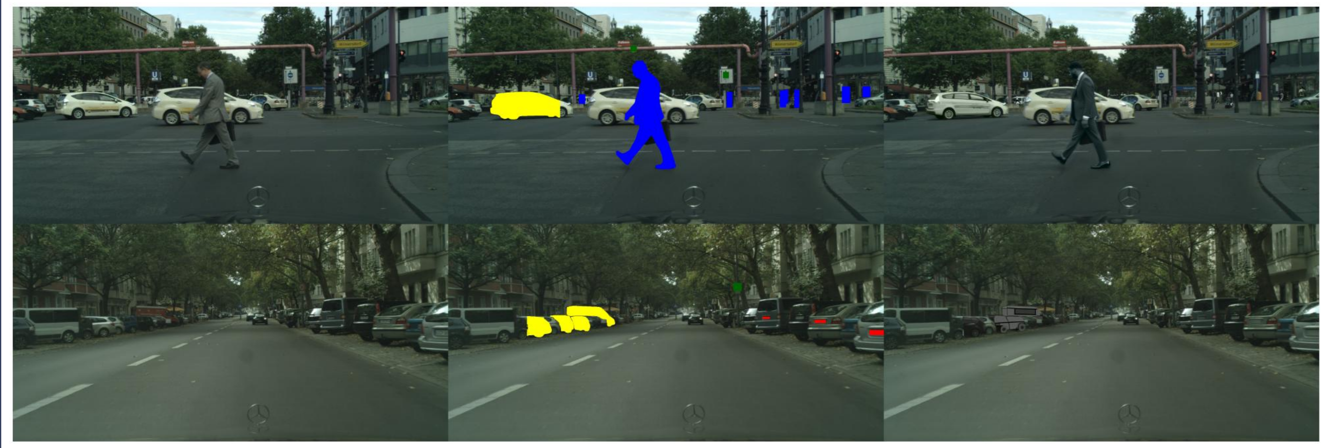

Fig. 4. Example pipeline output: left=original, center=segmentations (blue=person, red=license plate, green=traffic sign, yellow=indirect PII), right=anonymized. Subject image © CityScapes [18], research use.

Threat model

- Automated Re-ID at scale. ResNet-50 [19] with triplet [21] + center [22] loss trained on original CUHK03-NP-detected (767 IDs); attacker queries 1,400 originals against 5,332 anonymized gallery images with re-ranking [17].
- Out of scope: acquaintance recognition via retained pose, training-data leakage from generators (used as replaceable component, no fine-tuning), and human re-identification by determined observers.

Open challenges & limitations

Throughput. 133.5 s/img on dual L40S - suits offline curation, not real-time. **Hybrid architectures**. Zero-shot LVLMs underperform on pixel-precise localization of faces and bodies; routing direct PII to specialized detectors remains an open direction. **Ablation depth**. Pipeline-level ablation isolates agentic value; the influence of single components (diffusion backend, max. #rounds, IoU threshold) needs further research. **Audit reliability**. Trails support - not guarantee - GDPR transparency; L(V)LM hallucination is mitigated, not eliminated. Uncertain cases flagged for human review.

Conclusions

- Bounded multi-agent reasoning recovers open-vocabulary PII instances across categories that fixed-taxonomy detectors miss.
- Re-ID risk reduced with distribution fidelity.
- Downstream segmentation utility preserved, enabling continued use of anonymized datasets.
- Structured PDCA audit trails, dual-layer verification, and bounded iteration support GDPR transparency while flagging uncertain cases.

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JUST BETTER DATA

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